

Life under the ice

Although the water beneath the sea ice is cold, it is still much warmer than the ice itself. So, provided they can find food, many animals have no problem living under the ice. Tiny algae growing in the ice feed small animals that are hunted by fish and seals, and the seabed is often alive with sea urchins, starfish, and other marine invertebrates.

ICE GARDENS

The phytoplankton that produces nearly all the food in polar oceans does not grow in midwinter. It begins multiplying in early spring as the strengthening sunlight starts filtering through the thinning ice. Before long, the underside of the ice is covered with mats of green algae. These can be so thick that they stop the light reaching the deeper water below, and they provide vital food for the small creatures hunted by all the larger marine animals.



FROZEN FOOD

The tiny algae growing under the ice feed krill, copepods, and other types of zooplankton that have survived the winter by lying dormant (asleep) beneath the ice. As soon as the algae start multiplying, the animals start feeding intensively, and before long they are breeding too. Eventually, they will form vast swarms, but not until after the ice melts.

◀ ANTARCTIC KRILL

Green algae growing on the underside of the floating ice provides a half-frozen feast for these hungry krill in spring.

